

Learning Outcome/Objectives

LO₁: Understand the objectives of different software engineering phases (Chapter 1)

LO₂: Manage the software development process (Chapter 2)

LO₃: Analyse and model software requirements (Chapter 3)

Question 1: Multiple choice [20]

1. Which of the following is not a benefit of using Component-based software engineering in software development? [OBJ] [1]

- A) Improved software reusability
- B) Faster development cycles
- C) Enhanced maintainability
- D) Increased development complexity

Answer, LO₁ : D) Increased development complexity

2. What is the main purpose of the component-based software engineering? [OBJ] [1]

- A) To increase software development complexity.
- B) To reduce software reusability.
- C) To enhance software maintainability and reusability.
- D) To decrease software modularity.

Answer, LO₁ : C) To enhance software maintainability and reusability.

3. Which phase in the Waterfall Model involves creating detailed specifications for the software to be developed? [1]

- A) Requirements analysis
- B) Implementation
- B) Testing
- C) Design

Answer, LO₁ : A) Requirements analysis

4. What is the most accurate advantage of iterative development over the waterfall model? [1]

- A) It allows for late-stage changes and flexibility.
- B) It ensures all requirements are defined upfront.
- C) It requires less documentation.

D) It is faster and more efficient.

Answer, LO₁ : A) It allows for late-stage changes and flexibility.

5. What is a "black-box" component in Component-based Software Engineering? [1]

A) A component that is invisible and cannot be examined internally.

B) A component with no functionality.

C) A component that is tightly coupled with other components.

D) A component that is obsolete.

Answer, LO1: A) A component that is invisible and cannot be examined internally.

Chapter 2- [5]

A group of final year students are running a software engineering project. One of them is a project manager. Question 6-10 consist of questions directed to the project manager.

6. What is the purpose of a project charter in software project management? [1]

A) To define the software project's goals, scope, and stakeholders.

B) To write code.

C) To conduct quality assurance.

D) To create project documentation.

Answer: LO2: A) To define the project's goals, scope, and stakeholders.

7. What is the primary responsibility of a software project manager? [1]

A) Writing code and designing software.

B) Ensuring that the project is completed on time, within budget, and with the required quality.

C) Conducting user testing.

D) Creating project documentation.

Answer, LO2: B) Ensuring that the project is completed on time, within budget, and with the required quality.

8. Which of the following is NOT a primary constraint in software project management? [1]

A) Scope

B) Quality

C) Cost

D) Team size

Answer, LO2: D) Team size

9. Which project management tool is commonly used for scheduling and tracking tasks in a software project? [1]

A) Gantt chart

B) Microsoft Excel

C) Mind mapping software

D) Adobe Photoshop

Answer, LO2: A) Gantt chart

10. What is the purpose of a software project's risk assessment? [1]

A) To create a project schedule

B) To allocate resources to the project

C) To define the project's scope

D) To identify potential risks and their impact on the project

Answer, LO2: D) To identify potential risks and their impact on the project

11. Which phase is not concerned with assessing whether the system requirements are useful or not. [1]

A) Business (feasibility study);

B) Discovering requirements (elicitation and analysis)

C) Converting these requirements into some standard form (specification)

D) Use AI aligned technologies

Answer, LO3: D) Use AI aligned technologies

12. Which of the following is NOT a type of software requirement? [1]

A) Functional requirement

B) Non-functional requirement

C) User requirement

D) Coding requirement

Answer, LO3: D) Coding requirement

13. Which technique is commonly used for gathering requirements by observing and documenting how users currently perform their tasks? [1]

- A) Interviews
- B) Surveys
- C) Use cases
- D) Observational studies

Answer, LO3: d) Observational studies

14. What is the purpose of a use case in requirements analysis? [1]

- A) To describe how the software will be marketed
- B) To outline the software's technical architecture
- C) To specify the coding standards to be used
- D) To provide a detailed scenario of how a user interacts with the software

Answer, LO3: D) To provide a detailed scenario of how a user interacts with the software

15. Which of the following is an example of a non-functional requirement? [1]

- A) The system shall support more than 50 concurrent users.
- B) The response time for a user action should be less than 5 seconds.
- C) The software must be compatible with Ubuntu and macOS.
- d) The user interface must have a blue colour scheme.

Answer, LO3: B) The response time for a user action should be less than 2 seconds.

16. Which of the following is an example of an ambiguous requirement? [1]

- A) The system should be user-friendly.
- B) The system should have a login page.
- C) The system should be developed using Java.
- D) The system should perform at a speed of more than 500 transactions per second.

Answer, LO3: a) The system should be user-friendly.

17. Which of the following is NOT typically considered a stakeholder in software requirements analysis? [1]

- A) End users
- B) Developers
- C) Project managers
- D) Competitors

Answer, LO3: D) Competitors

18. What is the purpose of a Use Case diagram in software requirements analysis? [1]

- a) To show the sequence of tasks in a process
- B) To capture the interaction between actors and the system
- C) To list all the software requirements
- D) To create a Gantt chart for the project

Answer, LO3: B) To capture the interaction between actors and the system

19. Which technique is commonly used to prioritize requirements based on their importance to stakeholders? [1]

- A) Cost-benefit analysis
- B) Use case modeling
- C) Prototyping
- D) Code review

Answer, LO3: A) Cost-benefit analysis

20. Which type of requirement describes what the system should not do? [1]

- A) Functional requirement
- B) Non-functional requirement
- C) Constraint
- D) Use case

Answer, LO3: C) Constraint

Question 2: True or False [10]

Group 28, also known as My Easy Digital is a unified enterprise and high-tech solution. They help businesses with brand development by building their corporate identity from a blank page and integrating the brand into an online presence through e-Commerce-As-A-Service or Website Development. We further assist with hosting, website maintenance, and WordPress Security.

All questions are based on My Easy Digital company, state if statements are true or false.

1. The primary goal for this group is mainly on enterprise branding. [1]
Answer: True
2. Group 28 do provide both mobile and web applications [1]
Answer: True
3. WordPress security is regarded as an example of functionality requirement [1]
Answer: False
4. One of the risks associated with hosting websites is the availability [1]
Answer: True
5. If Group 28 aims to integrate the brand into an online presence, it means that this group will utilize waterfall model. [1]
Answer: False
6. Can Group 28 be capable of hosting podcasts on their space? [1]
Answer: True
7. My Easy Digital builds an online store on their platform according to developer's specifications. [1]
Answer: False
8. My Easy Digital provides hosting solutions, register customer's domain and provide a fixed method to create their emails. [1]
Answer: False
9. Part of managing projects, is to offer multiple website solutions through based on the user content. [1]
Answer: True
10. "We don't just do what you tell us, we engage you to deliver the best.", in software engineering, this phase is regarded as software project management. [1]
Answer: False

Question 3: Descriptive questions [20]

MTHETH SOFTWARE SOLUTIONS (MTSS) is a young and dynamic full stack software applications development company founded in 2017. MTSS is currently based in Gauteng, soon its footprint will be over the world. MTSS offers Android/IOS Mobile Apps, Web Applications, Information Security, Reporting Services, Legacy To Modern System and IT support.

1. Based on the services provided by MTSS, mention any potential client for Android/IOS Mobile Apps, Information Security and Reporting Services. [6]

Answer, LO1: Depends on the type of service the student will mention

2. Stakeholders often don't know what they want from the computer system except in the most general terms. They may find it difficult to articulate what they want the system to do or make unrealistic demands because they are unaware of the cost of their requests. Based on the client you mentioned in number 1 for Reporting Services, list any two functional requirements and two non-functional requirements. [8]

Answer, LO3: Will depend on what student mentioned in number 1

3. What are the risks associated with software of the stakeholder for Information Security you mentioned in number 1. [2]

Answer, LO2: Will depend on the kind of software student mentioned in number 1

4. It is hard to elicit domain knowledge during interviews for two reasons, mention those reasons [4]

Answer, LO3: Expression may vary from one student to another.

- A. All application specialists use terminology and jargon that is specific to a domain. It is impossible for them to discuss domain requirements without using this terminology. They normally use terminology in a precise and subtle way that is easy for requirements engineers to misunderstand.
- B. Some domain knowledge is so familiar to stakeholders that they either find it difficult to explain or they think it is so fundamental that it isn't worth mentioning. For example, for a librarian, it goes without saying that all acquisitions are catalogued before they are added to the library. However, this may not be obvious to the interviewer so it isn't taken into account in the requirements.

TOTAL MARKS: 50